

TASK3:

```
vennela_07@VennelaPC:~$ mkdir week3
mkdir: cannot create directory 'week3': File exists
vennela_07@VennelaPC:~$ cd week3
vennela_07@VennelaPC:~/week3$ pwd
/home/vennela_07/week3
vennela_07@VennelaPC:~/week3$ nano task3.c
vennela_07@VennelaPC:~/week3$ gcc task3.c -o task3
vennela_07@VennelaPC:~/week3$ ls
File1.txt  process.c  task2      task3.c      task4.c
File2.txt  task1      task2.c    task3.txt    task5
file1.txt  task1.c    task3      task4        task5.c
vennela_07@VennelaPC:~/week3$ ./task3
Parent Process Started
Parent PID: 882
Parent PPID: 588

Parent Process
Child PID: 883
Parent is Running...

Child Process Started
Child PID: 883
Child PPID: 882
Child is Running...
Child Process Terminated
Child has finished.
Parent Process Terminated
```

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```
GNU nano 7.2 task3.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());
    printf("Parent PPID: %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if (pid == 0)
    {
        // Child process
        printf("\nChild Process Started\n");
        printf("Child PID: %d\n", getpid());
        printf("Child PPID: %d\n", getppid());

        printf("Child is Running...\n");

        sleep(5);

        printf("Child Process Terminated\n");
    }
    else
    {
        // Parent process
        printf("\nParent Process\n");
        printf("Child PID: %d\n", pid);
        printf("Parent is Running...\n");

        wait(NULL);

        printf("Child has finished.\n");
        printf("Parent Process Terminated\n");
    }

    return 0;
}
```

TASK4:

```
vennela_07@VennelaPC:~/week3$ nano practicaltask4.c
vennela_07@VennelaPC:~/week3$ gcc practicaltask4.c -o practicaltask4
vennela_07@VennelaPC:~/week3$ ./practicaltask4
Process started
PID: 934
Process is Running...
Process is Running...
Process is Running...
Process is Running...
Process is Running...
Process is Running...
Process is Running...
^C
vennela_07@VennelaPC:~/week3$ ps -p 2500 -o pid,ppid,state,cmd
  PID  PPID  S  CMD
vennela_07@VennelaPC:~/week3$ top
top - 04:00:18 up 37 min, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7767.6 total,  7097.9 free,   515.1 used,   309.2 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,  7252.5 avail Mem

  PID USER  PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root    20   0  21720 13072 9732  S   0.0   0.2   0:00.77 systemd
    2 root    20   0   3180  2204 2072  S   0.0   0.0   0:00.02 init-systemd(Ub
    6 root    20   0   3196  2132 2008  S   0.0   0.0   0:00.00 init
   52 root    19  -1  66828 15920 14704  S   0.0   0.2   0:00.44 systemd-journal
  101 root    20   0  25404  6836  5168  S   0.0   0.1   0:00.46 systemd-udev
  111 systemd+ 20   0  21336 13132 11008  S   0.0   0.2   0:00.06 systemd-resolve
  122 systemd+ 20   0  91028  7964  6992  S   0.0   0.1   0:00.10 systemd-timesyn
  165 root    20   0   4244  2676  2420  S   0.0   0.0   0:00.02 cron
  166 message+ 20   0   9644  5380  4672  S   0.0   0.1   0:00.09 dbus-daemon
  195 root    20   0  17972  8884  7836  S   0.0   0.1   0:00.07 systemd-logind
  200 root    20   0 1756104 10996  8692  S   0.0   0.1   0:00.21 wsl-pro-service
  225 syslog  20   0  22516  6052  4628  S   0.0   0.1   0:00.06 rsyslogd
  244 root    20   0   3124  1972  1836  S   0.0   0.0   0:00.00 agetty
  259 root    20   0 107024 23200 13740  S   0.0   0.3   0:00.09 unattended-upgr
  344 root    20   0   6696  4668  3888  S   0.0   0.1   0:00.00 login
  399 vennela+ 20   0  20108 11372 9448  S   0.0   0.1   0:00.08 systemd
  400 vennela+ 20   0  21156  3596  1872  S   0.0   0.0   0:00.00 (sd-pam)
  416 vennela+ 20   0   6080  5384  3708  S   0.0   0.1   0:00.04 bash
  584 root    20   0   3188  1116   980  S   0.0   0.0   0:00.00 SessionLeader
  586 root    20   0   3204  1256  1108  S   0.0   0.0   0:00.35 Relay(588)
  588 vennela+ 20   0   6080  5400  3660  S   0.0   0.1   0:00.04 bash
  942 vennela+ 20   0   9400  5744  3508  R   0.0   0.1   0:00.00 top
```

```
GNU nano 7.2 practicaltask4.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process started\n");
    printf("PID: %d\n", getpid());

    while (1)
    {
        printf("Process is Running...\n");
        sleep(2);
    }

    return 0;
}
```

